Yunzhe Hu

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Education

The University of Hong Kong — Hong Kong, China
Ph.D. in Computer Science (Advisor: Prof. Dong Xu) — *Sept. 2022 – June 2027 (Expected)*

Shanghai Jiao Tong University — Shanghai, China
B.S. in Control Engineering (Artificial Intelligence Track) — *Sept. 2018 – June 2022*

Publications and Preprints (* Equal contribution)

- **Yunzhe Hu**, Dong Xu. VPFlow: Value-Predictive Planning with Imperfect World Models for Offline Goal Reaching. *Under review*, 2026.
- **Yunzhe Hu**, Difan Zou, and Dong Xu. Hyper-SET: Designing Transformers via Hyperspherical Energy Minimization. *ICLR*, 2026.
- **Yunzhe Hu**, Difan Zou, and Dong Xu. An In-depth Investigation of Sparse Rate Reduction in Transformer-like Models. *NeurIPS*, 2024.
- Ming Yang, **Yunzhe Hu**, et al. Complexity-Aware Policy via Heterogeneous Experts for Robotic Manipulation. *IEEE TCSVT*, 2025.
- Guangming Wang*, **Yunzhe Hu***, et al. What Matters for 3D Scene Flow Network. *ECCV*, 2022.
- Guangming Wang*, **Yunzhe Hu***, et al. Residual 3-D Scene Flow Learning With Context-Aware Feature Extraction. *IEEE TIM*, 2022.

Experience

Graduate Research Student — Sept. 2022 – Present
The University of Hong Kong (Advisor: Prof. Dong Xu) — *Hong Kong, China*

- Demonstrated that accurate next-state prediction is unnecessary for effective planning in offline goal reaching, challenging a pervasive assumption; developed *VPFlow* using closed-form hindsight returns to learn value-predictive world models without state reconstruction or bootstrapped targets.
- Identified off-distribution action search as a planning failure mode; designed noise-space evolutionary search for flow policy with imagined advantages to improve goal reaching at test time without weight updates, achieving mean success rate gains of at least 10.1 percentage points on locomotion and manipulation tasks from OGBench.
- Formulated *Hyper-SET*, a principled Transformer design framework, from a maximum-likelihood principle that balances semantic alignment and subspace uniformity, revealing a previously unnoticed connection; derived symmetric attention, feedforward layers, and normalization purely through constrained energy minimization, yielding interpretable recurrent Transformers for reasoning, classification, and masked modeling (*ICLR 2026*).
- Identified theoretical pitfalls in constructing attention layers from sparse rate reduction; derived model variants and linked sparse rate reduction to generalization through empirical complexity analysis, yielding improved regularization within the resulting Transformer family (*NeurIPS 2024*).

Undergraduate Research Assistant — Dec. 2020 – June 2022
Shanghai Jiao Tong University (Advisor: Prof. Hesheng Wang) — *Shanghai, China*

- Introduced context-aware feature pyramids and explicit residual refinement to resolve correspondence ambiguities from large motion and repetitive patterns in point clouds, reducing scene flow estimation error by at least 25% (*IEEE TIM 2022*).
- Designed all-to-all flow embedding with bidirectional reliability validation; benchmarked correspondence and refinement design choices, achieving state-of-the-art scene flow estimation accuracy (*ECCV 2022*).

Software Engineering Intern — July 2021 – Aug. 2021
Shanghai Intelligent and Connected Vehicle R&D Center — *Shanghai, China*

- Built a CarSim–Simulink simulation environment and Python vehicle controller to prototype deep reinforcement learning for autonomous parking with OpenAI Gym.

Technical Skills

Languages: Python, C++, Bash **Frameworks**: PyTorch, JAX, NumPy, Hugging Face, ROS, MuJoCo
Distributed Training: PyTorch DDP/FSDP (up to 8 NVIDIA A100 GPUs)
Tools: Linux, Git, Docker, Weights & Biases, Jupyter Notebook, GitHub Actions